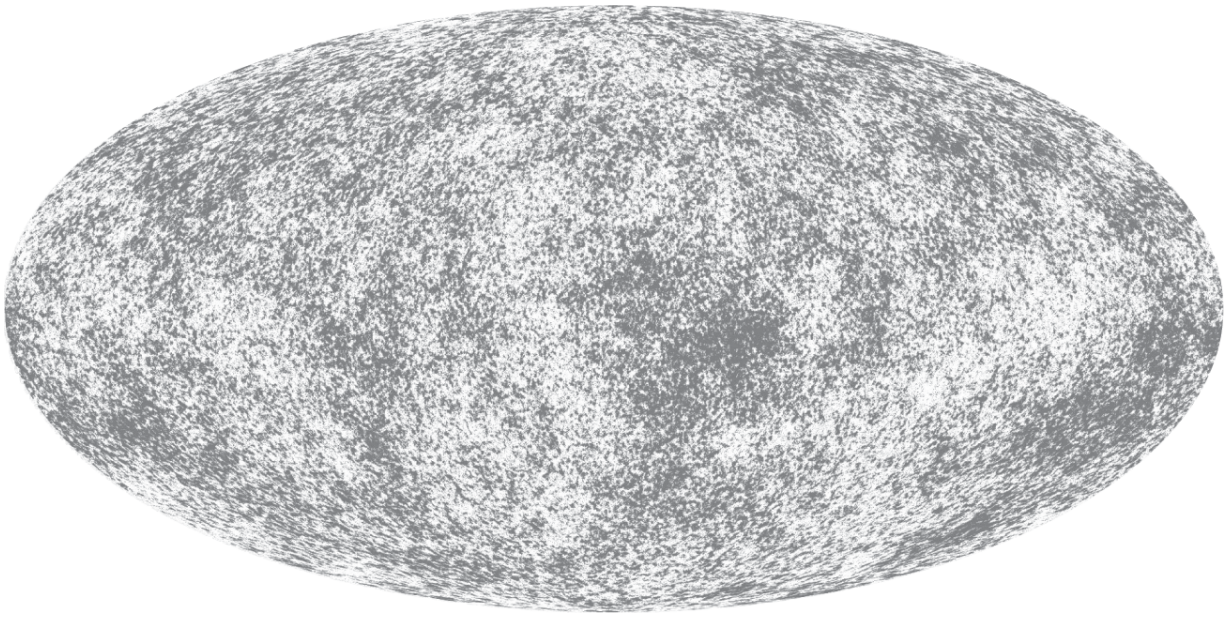


Figure 5: The cosmic microwave background. This is a microwave-frequency map of the whole sky projected onto an oval in a Mollweide projection (with the emission from our own galaxy removed). The darker regions indicate slightly cooler (lower frequency, or redder) microwave emission and the light regions are slightly hotter (higher frequency, or bluer) emission. These indicate, respectively, parts of the early universe that were slightly denser or slightly less dense than their surroundings, at a level of one part in 100,000.

So. The Big Bang happened. We've seen it, we've calculated it, the physics adds up. Now, let's all gather around in the glow of the cosmic blackbody and tell the origin story of the cosmos.

IN THE BEGINNING

Not all of cosmic history is as directly visible as the cosmic microwave background. The few hundred thousand years before the end of the fireball stage, and the half million or so years right after, are extremely hard to observe. In the former case, it's because of too much light (imagine trying to look through a wall of fire) and in the latter, because of too little (imagine trying to look at some specks of dust in the air between you and a wall of fire). But the CMB, right in the middle, gives us a solid anchor to extrapolate from in each direction, and now we have a compelling narrative for how the universe evolved over time, starting from the first billionth of a billionth of a billionth of a second and arriving at today, 13.8 billion years later.

Shall we?